

Thumb Force Estimation with Egocentric Vision

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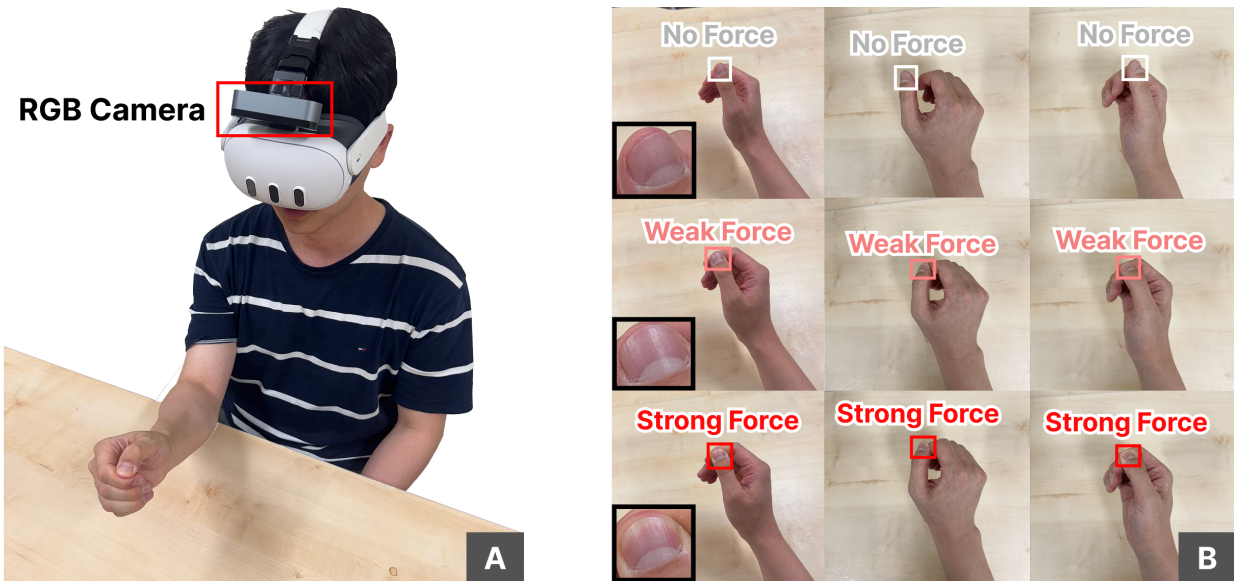


Figure 1: (A) The proposed sensing system captures the hand from an egocentric viewpoint using a single RGB camera attached to an HMD, and (B) estimates the location and force of the thumb in real-time based on a deep learning model to infer three force classes (No Force, Weak Force, Strong Force).

Abstract

Thumb-based touch interaction plays a key role in enhancing usability in virtual and augmented environments. Existing vision-based force sensing approaches rely on hand localization. In this work, we propose a novel force sensing system that simultaneously detects the location and force of thumb. This system demonstrates robust thumb's force detection performance for subtle gestures in egocentric view.

CCS Concepts

• Human-centered computing → Gestural input; • Computer systems organization → Real-time operating systems; • Computing methodologies → Activity recognition and understanding.

*Both authors contributed equally to this research.

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Keywords

Computer Vision, Thumb Force, Egocentric, Sensing

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1 Introduction

Thumb-based touch interaction is pivotal for enhancing usability in virtual and augmented reality (AR/VR). The use of bare-hand input improves usability by supporting natural interactions without the burden of additional hardware [3, 5]. Previous research has attempted to detect hand contact force using glove-based systems [2], but issues such as discomfort, high cost, and environmental constraints have made them unsuitable for AR/VR.

Recently, vision-based hand force estimation methods using a single RGB camera have recently gained attention [1, 7]. However, most existing methods use a two-step pipeline structure in which the hand is first detected using an external localization module [4], followed by a force estimation based on the detected area. This

approach is vulnerable to detection errors and prone to latency of inference, making the system unsuitable for real-time systems.

In this demo, we present a new thumb force sensing system that is a localization-free end-to-end model. This model predicts the location and contact force simultaneously using an egocentric RGB frame. Since our system operates without separate hand localization module, inference speed is fast and exhibits robust performance even under hand occlusion and lighting changes. Our system enables force-based bare-handed input in various scenarios, from subtle gestures to interactions in real-time.

2 Implementation

Our thumb force sensing system consists of an RGB camera, which is attached to the top of a head mounted display (HMD). This camera is connected to a computer via USB and captures hands from an egocentric view at a speed of 30 frames per second (30Hz). The captured frames are stored individually and processed using a PyTorch-based deep learning model. Our implementation leverages RT-DETRv2 [6] model. This model simultaneously infers the location and contact force of the thumb, minimizing latency and improving robustness. We designed our model to classify three classes of force: No Force (force not activated), Weak Force (1 N or less), and Strong Force (5 N or more).

We utilized 3.3 million hand interaction frames collected from 41 participants, using data from 33 participants for training and data from 8 participants for testing. The experimental results achieved a force classification accuracy of 93.46%.

3 Demo Application

Our system enable to infer and visualize hand force in real-time using only a single RGB camera, and we use this to extend bare-handed input interaction. In this demo, we demonstrate the real-time performance and applicability of the proposed system.

3.1 Real-time Thumb Force Visualization Interface

This demo shows the proposed system's ability to detect and visualize the force and location of the thumb visible to the camera in real-time. When the user presses their thumb to the index finger in the camera view, the screen displays the location of the thumb finger in a blue box, along with the detected force class (No Force, Weak Force, Strong Force) in text. This system provides real-time video streams and prediction results intuitively through an OpenCV-based user interface (UI). This allows users to receive immediate feedback on their hand movements and the corresponding force recognition results, enabling them to directly verify the real-time performance and precision.

3.2 Dinosaur Jumping Game

We developed an enjoyable jumping game interface using the force detection results of the proposed system. The user can input jump commands using only the force applied to their thumb, depending on the magnitude of the input force, the dinosaur in the game will jump to avoid the tree or remain still. For example, a weak force results in a light jump, a strong force results in a high jump, and no force results in no jump. This intuitively demonstrates

how thumb force-based interaction can be utilized as actual input events, suggesting that the proposed sensing system is sufficiently applicable even with a simple bare-handed gesture-based interface.

3.3 VR Interface Navigation and Manipulation

The third demo showcases how the proposed thumb force sensing system can be applied to interface control interaction and drag control tasks in VR environment. Within the VR interface, the user can manipulate UI elements using only bare-handed gestures captured from an egocentric camera. Crucially, the behavior of the system dynamically adapts to the magnitude of the thumb force applied on the index finger. When a weak force is detected, the system interprets this as a simple touch input. This allows users to interact with discrete elements, such as toggling buttons or changing display channels on a virtual control panel. In contrast, a strong force triggers a drag gesture, enabling more powerful operations such as selecting and moving multiple objects simultaneously or toggling the virtual display's power state.

4 Conclusion and Future Work

The proposed system is a vision-based thumb force sensing system that simultaneously predicts location and contact force using an egocentric single RGB frame. This system demonstrates the potential for accurate and intuitive force-based hand input through demo applications such as real-time visualization and game control. In future work, we will explore the applicability of the proposed system by extending it to a wider range of interactive environments and integrating it into practical application scenarios such as object and 3D interaction, multi-finger input, and UI control in VR.

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